Review Questions for Lecture 16

1. What does the Glivenko-Cantelli Theorem say?

2. What does the Central Limit Theorem say?

3. Explain briefly why it is not always possible to calculate probability histograms exactly.

4. Give two different methods to approximate probability histograms. Explain briefly under what circumstances it is possible to use each method.

5. Consider a super lopsided box with 99 0's and 1 1. Use simulation to create approximate probability histograms for the percentage of 1's for $n = 2, n = 10, n = 100, n = 1{,}000$, and $n = 10{,}000$ draws with replacement. At what point is the histogram roughly bell-shaped?

6. Repeat question (5) for the mean of n draws with replacement from a box containing $\{-15,0,0,0,0,2,2,2,2\}$.

7. Calculate the exact probability that the sum of three draws with replacement from the box {1,2,3} will be equal to 6. (Hint: There are $3^3 = 27$ possible outcomes. What are they? And how many of result in a sum of 6?)

8. Write some R code that simulates drawing from the box {1,2,3} three times with replacement and computing the sum.

9. Using your code to question (8), write additional R code that estimates the probability that the sum will equal 6. Use 10,000 simulations.

10. Using the Central Limit Theorem, approximate the probability that the sum of three draws (with replacement) from the box {1,2,3} will be equal to 6.

11. Compare your answer to question (10) to your answer to question (7). Comment on the quality of the normal approximation, keeping in mind that there were only three draws.

12. How many draws are necessary for the normal approximation to be any good? Or is not possible to say? Explain briefly.

13. What are the advantages of approximating probabilities using computer simulation vs. using the normal approximation?